

## PERSPECTIVE

# Pursuing Clinical Trials “Adaptable” to the Moment: Advancing Relevance, Reach, and Rapid Learning With Rigor

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**T**he speed of science is accelerating. Genomics, imaging, computational biology, digital health technologies, artificial intelligence, and biotechnology are compressing the path from discovery to translation and expanding what is possible to diagnose, prevent, and treat cardiovascular disease.<sup>1</sup> Yet the speed of clinical evidence has not kept pace.<sup>2</sup> Trials that determine whether discoveries improve health remain slow, costly, complex, and disconnected from the settings where people live or receive care. In the digital age, evidence generation and implementation should increasingly occur together across patients, clinicians, health systems, and communities so that learning can happen faster and everywhere evidence is needed.<sup>3</sup> As an editorial group at *Circulation*, we see this gap as both an obligation and an opportunity to encourage submissions that advance cardiovascular health with relevance, reach, and rapid learning with rigor. This includes primary trial results, high-impact secondary analyses that extend landmark studies, target-emulated trials, and methods, or, additionally, perspectives, including those related to regulatory policy that change or challenge the status quo of burdensome and untimely evidence generation. In the sections below, the following themes guide clinical trial submissions that we hope to publish in *Circulation*.

## RELEVANCE

We are especially interested in research that addresses decisions faced by patients, clinicians, health systems, regulators, payers, and policymakers. The traditional separation between the world of trials and the world of patient care is increasingly untenable as digital tools, virtual

care, health-system data, and digital health data create new opportunities to study interventions in routine settings. In the pages of *Circulation*, leaders have noted that ethical and scientific standards can be preserved with new methods that meet patients where they live, which may overcome the cost, complexity, and delay of traditional clinical trials.<sup>4</sup> Relevant submissions should explain why the question matters now, how the design reflects contemporary care, and how the findings could influence treatment selection, safety, symptoms, quality of life, implementation, policy, or future research. Secondary analyses are strongest when prespecified or scientifically compelling and able to clarify heterogeneity of treatment effect, mechanisms, patient-reported outcomes, safety, implementation, or health-system implications.

## REACH

We encourage submissions that extend evidence to the populations and settings in which cardiovascular disease occurs. Whether this is addressing a simple question or a more complex healthcare issue, we hope that clinical trials can be more adaptable in their reach to engage and conduct research where its most convenient, including in the home.<sup>5</sup> This is especially important for prevention, for which outcomes are shaped by risk factor control, access, health literacy, transportation, food and housing insecurity, financial strain, and other social determinants of health. Decentralized and hybrid approaches can reduce burden through direct-to-participant recruitment, electronic health record-enabled procedures from identification to data collection, remote screening and consent,

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home delivery of interventions, digital health or telehealth technologies for patient-reported outcomes, wearable sensors, pharmacy distribution, local testing, and registry or claims-based outcomes.<sup>6</sup> For both primary reports and secondary analyses, authors should consider whether evidence is applicable across geography, race and ethnicity, sex, age, socioeconomic circumstances, comorbidities, digital access, and care settings.

## RAPID LEARNING

Although we are far from the ideals of a learning health-care system, we welcome submissions that help clinical evidence move closer to the pace of discovery while maintaining trustworthiness. Likewise, as an editorial group, we aim to help the community rapidly learn from trial results through packages that connect discovery science to implementation through rapid, far-reaching channels to patients, clinicians, and key stakeholders. Platform, adaptive, pragmatic, and digitally enabled trials can turn unanswered questions into learning opportunities embedded in care or everyday life. Recent examples include shared infrastructure for multiple interventions, bayesian analyses with prespecified rules, shared or concurrent control groups, response-adaptive randomization, and master protocols that add new questions as science evolves.<sup>7</sup> As artificial intelligence moves from theory to application in clinical and trial settings, we are interested in evaluations of artificial intelligence-enabled approaches that may rapidly improve the conduct of clinical trials from engagement to recruitment, analysis, and dissemination. We also welcome studies that test whether artificial intelligence tools can rapidly improve cardiovascular health and well-being across diverse health settings. These methods will be judged by the same standards as other trial innovations, ensuring transparency, validation, attention to bias, and evidence that they improve efficiency without compromising trust or unfavorably altering benefits to risks.

## RIGOR

Relevance, reach, and rapid learning will matter only if the evidence remains trustworthy. We are fortunate that the demanding culture of the cardiovascular community is held up as the gold standard, which has made cardiovascular medicine evidence based with the highest standards. In *Circulation* >2 decades ago, Califf and DeMets<sup>8,9</sup> emphasized that randomized trials remain essential because biologic plausibility, expert opinion, and early experience can be misleading. The cardiovascular trials enterprise has also shown, through sustained collaboration, multicenter infrastructure, and methodological innovation, how rigorous evidence can generate decades of progress in care.<sup>10</sup> Those lessons remain essential as randomized trials continue to evolve. However adaptable the design or mod-

ern the method, the central questions remain unchanged: Were the correct patients (or individuals) studied? Was the appropriate intervention (or exposure) tested or observed? Were the optimal efficacy and safety outcomes evaluated? Was Good Clinical Practice maintained? As clinical research evolves with pragmatic approaches, platform trials, digital tools, and modern analytic methods, rigor must be maintained through clear hypotheses, credible end points, data quality, patient safety, transparency, protection from bias, appropriate monitoring, reproducibility, responsible communication, and valid inference.

## WHAT AND HOW WE HOPE TO PUBLISH

*Circulation* aims to be a global home for high-quality cardiovascular trials across phases and designs, including traditional randomized trials, early- to late-stage research to implementation, pragmatic and adaptive trials, platform trials, embedded healthcare trials, and carefully conducted target trial-emulated studies using high-quality data and appropriate causal methods. We welcome primary results that can inform practice, policy, or future research and secondary analyses of landmark trials that address important questions not fully answered by the primary report. We also encourage viewpoints that clarify an important moment for the field or propose ideas that may change the status quo or improve regulatory science. Our editorial approach will honor your work and all of the participants. We will focus on the importance of the question, the strength of the design, the credibility of the data, and the potential clinical impact, not simply whether results are positive, while avoiding the classic endless review cycles that every one of us has experienced.

## CONCLUSIONS

Cardiovascular medicine needs clinical research that is easily adaptable to the moment through its relevance, reach, rigor, and how we, together, can achieve rapid learning. Through the pages of *Circulation* and beyond, we aim to close the gap between scientific discovery and clinical evidence and to accelerate better cardiovascular health.

## ARTICLE INFORMATION

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### Disclosures

None.

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